BijliCast - Dataset Proposal

HackInsight '26 · Energy demand track · Problem: hourly demand planning for K-Electric (Karachi) and Pakistan's nine DISCOs. All counts below are generated from the warehouse and manifest.csv by `python -m bijlicast.publish proposal`.

1. Sources (all credible, original publishers, gathered during the event)

Source	Publisher	What we use	Access / licence	Update	Volume
ISMO DISCO & K-Electric monthly workbooks	Independent System and Market Operator (ISMO), Government of Pakistan	Average demand for each hour of the day (MW) for K-Electric + 9 DISCOs; monthly energy; generation mix by hour	Public download links on ismo.gov.pk/dashboard; robots.txt allows all; attribution to ISMO	Monthly (~15 days after month end)	39 workbooks; 7,200 hourly-profile rows (28 usable months Sep 2023-Sep 2026); 5,040 generation-mix rows
ISMO hourly marginal price	Independent System and Market Operator (ISMO), Government of Pakistan	System marginal price (Rs/kWh) per hour	Same as above	Monthly (cumulative file)	19,728 hourly rows (Jul 2024-Sep 2026)
ERA5 reanalysis via Open-Meteo (+ live forecast)	Open-Meteo.com (ERA5 / ERA5-Land reanalysis from Copernicus Climate Change Service, ECMWF; forecasts from national weather services)	Hourly temperature, humidity, dew point, apparent temperature, radiation, cloud, wind, rain at each DISCO's headquarters city	CC BY 4.0 (Open-Meteo data); ERA5 under the Copernicus licence. Free non-commercial API, no key.	Hourly (forecast) / daily (archive)	10 cities x 2022-01-01..2026-10-12 (UTC) = 419,040 hourly rows
NEPRA State of Industry Report 2024	National Electric Power Regulatory Authority (NEPRA)	Trusted text for the Insight Agent's citations	Public regulatory publication (nepra.org.pk); robots.txt allows all. Cited with attribution.	As published	1 document
Indicative Generation Capacity Expansion Plan 2025-35 (Rev. 2)	Independent System and Market Operator (ISMO), filed with NEPRA	Trusted text for the Insight Agent's citations	Public regulatory filing (nepra.org.pk); robots.txt allows all. Cited with attribution.	As published	1 document
K-Electric Tariff Structure (Time-of-Use peak hours, FCA, customer contacts)	K-Electric Limited	Trusted text for the Insight Agent's citations	Public web page (ke.com.pk); robots.txt disallows only /wp-admin/. Cited with attribution.	As published	1 document

Total volume: 452,075 rows across independent publishers (minimum required: 50,000), plus 3 official documents. Every raw file is in manifest.csv (source, query URL, fetch time, SHA-256, licence); `make data` re-downloads and verifies them from an empty folder.

2. Which source is not on the track list, and why it is credible

The Energy track lists ENTSO-E, Open-Meteo, UCI Electricity Load Diagrams and ETT. **ISMO** and **NEPRA** are not on it. ISMO is the Government of Pakistan's Independent System and Market Operator (formerly part of NTDC/CPPA-G); it runs the national grid's market and publishes the data from the National Grid Company's metering service provider. NEPRA is the national power regulator. Both are original publishers, not re-uploads. ISMO and Open-Meteo are fully independent: grid metering versus satellite/model reanalysis.

3. Why this data fits the problem and the primary user

K-Electric's planning desk decides how much to draw from the national grid for each hour of the coming month. ISMO's K-Electric figure is exactly that quantity, measured at the grid interface, and the hourly marginal price shows what each hour costs the system. Weather explains cooling-driven load, and NEPRA/ISMO/KE documents give the policy context (peak hours, losses, demand scenarios) the agent cites. The data reflects Pakistan throughout: its DISCOs, provinces, Ramadan load pattern (sehri spikes found by the Data Quality Agent), summer cooling peaks and the 2026 fuel crisis.

4. Declared baseline (fixed before results; committed to git first)

Seasonal naive: demand for entity e, month m, hour h = demand in the same month last year at the same hour; if unpublished, the latest month already published at forecast time (≤ m-2). Time-based split; test months Jan-Sep 2026; success = at least 10% lower MAPE and K-Electric better. See docs/baseline_declaration.md.

5. What the Data Quality Agent found in the official data (latest run)

Check	Action	Decisions
DQ-ISMO-STALE	FLAG	18
DQ-ISMO-CONSIST	FLAG	8

Check	Action	Decisions
DQ-ISMO-SPIKE	FLAG	8
DQ-ISMO-NOPROFILE	FLAG	5
DQ-ISMO-UNIT	FIX	4
DQ-SRC-MISSING	FLAG	3
DQ-ISMO-COPY	FLAG	3
DQ-ISMO-AGG	FLAG	2
DQ-ISMO-DUP	FIX	2
DQ-ISMO-CONSIST-SUMMARY	FLAG	1
DQ-ISMO-GAPS	FLAG	1
DQ-ISMO-UNIT	FLAG	1

Highlights: whole months that ISMO re-uploaded as copies of the previous month (Jul 2024, Apr 2025) are excluded; K-Electric energy values repeated for 6+ months are flagged as stale; broken publisher links (Jun-Jul 2026) are flagged, not imputed.

6. Local validation

ISMO's K-Electric profile for March 2024 averages **944 MW** (rows DP-KE-202403-00..23). Dawn reported that K-Electric draws 1,000-1,100 MW from the national grid at peak (dawn.com/news/2036015), which is consistent with ISMO measuring KE's grid draw rather than its total demand. K-Electric's own tariff page (DOC-KETOU) confirms the peak window we use (6-10 PM).

7. Not used

No Kaggle/GitHub re-uploads, no synthetic or AI-generated data for training or evaluation, no personal data. LUMS household smart-meter data (REWD-P) was considered and left out because its licence is not stated.